**Metadata: Shrub-grass dynamics - 20250716**

Rachel R. Renne, Yale School of the Environment

This document contains a description of the methods used to collect data for sites visited between 2021 and 2023. Descriptions are ordered alphabetically according to the name of the data files. Tables at the end of each section provide an explanation of the fields in each data file. All data may not have been collected at all sites.

**codes_withpathway.csv**

Information about plant and other codes found in the data. Functional type codes include: annual forb (AF), annual grass (AG), big sagebrush (ARTR), bedrock (BR), boulder (BY), C3 annual grass (CAG), cobble (CB), C3 perennial bunch grass (CBG), C3 perennial rhizomatous grass (CRG), embedded litter (E), gravel (GR), herbaceous litter (HL), lichen crust (LC), moss (M), None (N), perennial forb (PF), perennial grass (PG), Sandburg's bluegrass (POSE), perennial sedge (PSE), soil (S), spike moss (SEL), shrub (SH), subshrub (SSH), stone (ST), succulent (SUC), tree (TR), unknown (UNK), vagrant lichen (VL), C4 annual grass (WAG), C4 perennial bunch grass (WBG), woody litter (WL), and C4 perennial rhizomatous grass (WRG). Growth form codes are broader categories.

| Field name | Description |
|---|---|
| code | Code as found in the data |
| scientificname | Scientific name corresponding to the code |
| commonname | Common name associated with the code |
| family | Family for each plant species |
| FunctionalType | Detailed functional types including duration and photosynthetic pathway |
| GrowthForm | Growth form codes for each plant (broader categories) |

**community.csv**

At each target shrub on the plot, I placed one 20 x 30-cm quadrat at the base of the target shrub stem with the 20 cm length parallel to a line between the target shrub stem and the neighboring shrub stem and centered over the line (15 cm on each side) and another quadrat centered on the line halfway between the target shrub and its neighbor. If the target shrub had no suitable neighbor, I placed a quadrat approximately centered on the shrub stem and with 100% canopy cover. In each quadrat, I recorded the size and number of all herbaceous and succulent plants that were at least 50% rooted within quadrat. For bunch grasses, I recorded species, maximum basal width of the bunch, the basal width perpendicular to this measurement, the average maximum height of leaves, and number of tillers estimated to nearest 10. For perennial forbs, I recorded species, maximum width, the width perpendicular to this measurement, and the height. For rhizomatous grasses, annual grasses, and annual forbs, I recorded species and height.

| Field name | Description |
|---|---|
| plot | Plot name |
| transect | Transect number (1-3) |
| microsite | "canopy" for the quadrat adjacent to the shrub stem and "interspace" for the quadrat situated between the target shrub and its neighbor |

| | |
|---|---|
| canopy | Canopy cover of each individual: 0 = open, 1 = 100% shrub canopy cover, 0.5 = mixed |
| point | Point in meters along the transect |
| species | USDA plants species code, functional type, or unknown. In 2021, I occasionally recorded functional type (AF = annual forb, PF = perennial forb) instead of species for forbs. NO PLANTS = no plants present in the quadrat, NO GAPS >60 cm = no interspace microsite for that point along the transect. |
| length | Maximum basal width of perennial grasses or maximum width of forbs (cm) |
| width | Width perpendicular to length (cm) |
| height | Average maximum height of perennial grass leaves or height of other functional types (cm) |
| number | Number of individuals (0 or 1) |

**CoreARTR_combined_DayMet_cropped&trimmed.tif**

Potential big sagebrush (*Artemisia tridentata*) habitat, as described in Renne et al., (2021).

**dung.csv**

I counted and recorded the age (recent or old) and species of all dung in 2 x 25-m belt transects centered around each of the three 25-m transects.

| Field name | Description |
|---|---|
| plot | Plot name |
| transect | Transect number (1-3) |
| animal | Species or animal type |
| age | Estimated age of dung (recent or old) |
| count | Count for each species on each transect |
| RRR | Final decision on how to handle error |

**lpi.csv**

Line-point intercept protocol followed Herrick et al., (2005).

| Field name | Description |
|---|---|
| plot | Plot name |
| transect | Transect number (1-3) |
| point | Point in meters along the transect |
| toplayer | Top species encountered by pin (USDA Plants code or unknown code for each species), N = No plants |
| layer1 | Second new species/object encountered by pin |
| layer2 | Third new species/object encountered by pin |
| layer3 | Fourth new species/object encountered by pin |
| layer4 | Fifth new species/object encountered by pin |
| soilsurface | Soil surface |
| toplayer_dead | Indicates if the part of the plant touching the pin in top layer is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |

| | |
|---|---|
| layer1_dead | Indicates if the part of the plant touching the pin in layer 1 is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |
| layer2_dead | Indicates if the part of the plant touching the pin in layer 2 is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |
| layer3_dead | Indicates if the part of the plant touching the pin in layer 3 is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |
| layer4_dead | Indicates if the part of the plant touching the pin in layer 4 is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |
| soilsurface_dead | Indicates if the part of the plant touching the pin at the soil surface is dead (1 = dead, 0 = live). If both live and dead parts of the same species touch the pin, the plant is recorded as live. |
| plantnotes | Notes on the plants encountered at this point on the line, generally a guess at the genus and/or species of any unknown plants for plots visited in 2021. |

**plots.csv**

Plots consist of three evenly spaced 25-m transects arranged in a spoke design following Herrick et al. (2005). If transects intersected large animal burrows or anthills, I shifted individual transects to avoid these features. I recorded slope (2022 and 2023), aspect, and slope shape at each plot.

| Field name | Description |
|---|---|
| plot | Plot name |
| latitude | Latitude of plot center recorded at plot (using GPS or iPad) |
| longitude | Longitude of plot center recorded at plot (using GPS or iPad) |
| elevation | Elevation (m) of plot center recorded at plot (using GPS or iPad) |
| date | Sampling date (YYYYMMDD) |
| augerdepth | Soil depth attained using auger |
| aspect | Aspect in degrees measured using a compass |
| percentslope | Percentage slope measured with a clinometer (2022/2023) or estimated using a digital elevation model (GP5-005; USGS, 2018) |
| slopeshape | Shape of slope looking downslope and perpendicular to this direction. L = linear, C = concave, V = convex |
| t1_azimuth | Direction (degrees) of transect 1 |
| t2_azimuth | Direction (degrees) of transect 2 |
| t3_azimuth | Direction (degrees) of transect 3 |
| dominantshrub | Code (see **codes_withpathway.csv**) for dominant sagebrush species on plot. |
| averagedistance | Average distance between shrubs at plot |
| avg_PET | Average annual potential evapotranspiration (mm), estimated from SOILWAT2 output (Schlaepfer and Murphy, 2023) |
| avg_trans_0_30cm | Average annual transpiration (1991-2020) from top 30 cm of soil, estimated from SOILWAT2 output (mm) |

| | |
|---|---|
| avg_trans_30_200cm | Average annual transpiration (1991-2020) from 30-200 cm depth of soil, estimated from SOILWAT2 output (mm) |
| avg_trans_0_200cm | Average annual transpiration (1991-2020) from 0-200 cm depth of soil, estimated from SOILWAT2 output (mm) |
| prop_shallow | Proportion of total transpiration from to 30 cm of soil |
| sand | Fractional sand content by weight in top 30 cm of soil |
| clay | Fractional clay content by weight in top 30 cm of soil |
| map | Mean annual precipitation (mm) calculated from 1991-2020 (Thornton et al., 2022). See “01_Summarize_vegetation&climate.R” for data extraction code |
| mat | Mean annual temperate (C) calculated from 1991-2020 (Thornton et al., 2022). See “01_Summarize_vegetation&climate.R” for data extraction code |
| Jan_ppt – Dec_ppt | Average monthly precipitation (1991-2020) estimated from DayMet (Thornton et al., 2022). See “01_Summarize_vegetation&climate.R” for data extraction code |
| Jan_tavg – Dec_tavg | Average monthly temperature (1991-2020) estimated from DayMet (Thornton et al., 2022). See “01_Summarize_vegetation&climate.R” for data extraction code |

**Shallow_soil_sensitivity_simulation_results.csv**

Simulation output from SOILWAT2 simulations (Schlaepfer and Murphy, 2023) for 51 sites across the western United States. Each site was simulated for 30 years (1991-2020) with weather data from DayMet (Thornton et al., 2022). Each site was simulated with site-specific soil and plant cover information. Here, we have summarized transpiration for different depths of the soil.

| **Field name** | **Description** |
|---|---|
| Plot | Plot name |
| avg_trans_X_Ycm | These fields show average (1991-2020) annual transpiration from X to Y cm in the soil. |
| prop_shallow_Y | Proportion of total transpiration from shallow soil layers, defined as depths from 0 to Y cm. |

**Vegetation_sensitivity_simulation_results.csv**

Simulation output from SOILWAT2 simulations (Schlaepfer and Murphy, 2023) for 51 sites across the western United States. Each site was simulated for 30 years (1991-2020) with weather data from DayMet (Thornton et al., 2022). Each site was simulated 51 times, with site-specific soil and plant cover information from each of the 51 sites based on field data.

| **Field name** | **Description** |
|---|---|
| Plot | Plot name |
| shrubs | Fractional shrub cover for this plot |
| c3pgrass | Fractional C3 perennial grass cover for this plot |
| c4pgrass | Fractional C4 perennial grass cover for this plot |
| agrass | Fractional annual grass cover for this plot |

| | |
|---|---|
| pforbs | Fractional perennial forb cover for this plot |
| bareground | Fractional cover of bare ground for this lot |
| Plot names | Each column labeled as a plot name corresponds to the proportion of total transpiration from shallow soil layers (0-30 cm) for the plot in that row parameterized with plant cover data from the plot listed as the column name |

**References**


Herrick, J.E., Van Zee, J., McCord, S.E., Courtright, E.M., Karl, J.W., Burkett, L.M., 2005. Monitoring manual for grassland, shrubland, and savanna ecosystems, 2nd ed. United States Department of Agriculture, Agricultural Research Station, Jordana Experimental Range, Las Cruces, N.M.

Renne, R., Schlaepfer, D., Palmquist, K., Lauenroth, W., Bradford, J., 2021. Estimating complex ecological variables at high resolution in heterogeneous terrain using multivariate matching algorithms.

Schlaepfer, D., Murphy, R., 2023. rSOILWAT2: An Ecohydrological Ecosystem-Scale Water Balance Simulation Model.

Thornton, M.M., Shrestha, R., Wei, Y., Thornton, P.E., Kao, S.-C., Wilson, B.E., 2022. Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4 R1 0 MB. https://doi.org/10.3334/ORNLDAAC/2129

USGS, 2018. USGS National Elevation Dataset (NED). United States Geological Survey, Earth Resource Observation and Science Center (EROS).